Abdulbasit Oyebamiji Oyetunji

AI/ML Engineer — Medical Imaging & Healthcare AI

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PROFESSIONAL SUMMARY

AI/ML Engineer and 5th-year Medical Student at the University of Ibadan, working at the intersection of clinical medicine, medical imaging, and applied artificial intelligence. Research focuses on brain-tumour segmentation and domain adaptation on the BraTS-Africa dataset, with systematic investigation of Mamba and attention-based architectures for medical imaging. Research accepted at MICCAI, ISMRM, and Deep Learning Indaba. Combines clinical training with end-to-end AI engineering (LLMs, RAG, agentic AI, deep learning) to build robust, context-aware systems for real-world and low-resource healthcare settings.

RESEARCH EXPERIENCE & PUBLICATIONS

Token Mixer Placement Matters: A Systematic Encoder–Decoder Ablation Study of Mamba for Brain Tumour Segmentation on BraTS-Africa 2026

MICCAI 2026 (MIRASOL Workshop), Strasbourg, France — Poster.
Africa in AI Affinity Group at NeurIPS 2026 — Accepted.
Research Lead. Led design and execution of a deep learning study comparing Mamba- and Transformer-based token mixers across encoder–decoder placements for 3D brain tumour segmentation; owned study conception, model development, experimental design, data analysis, and manuscript preparation.

Mamba or Attention? A Systematic Encoder–Decoder Study of Token Mixing for Brain Tumour Segmentation on BraTS-Africa 2026

Deep Learning Indaba, Lagos, Nigeria — Poster presentation.

Domain-Adaptive Transformer for Data-Efficient Glioma Segmentation in Sub-Saharan MRI 2026

ISMRM Annual Conference, Cape Town, South Africa — Poster. Co-author. Contributed to a domain-adaptive transformer model (SegFormer3D+) for glioma segmentation under domain shift, integrating radiomics, intensity harmonisation, and attention mechanisms, with validated performance gains on BraTS-Africa.

From Binary Labels to Stage-Aware Bounding Boxes: Evaluating Annotation Depth in Malaria Microscopy 2026

MICCAI 2026 (MIRASOL Workshop), Strasbourg, France — Poster presentation.

Development of a Structured African Cardiovascular Disease Clinical Dataset for Precision Machine Learning and Risk Prediction Dec 2025 – Present

Principal Investigator. Leading development of a structured, large-scale African cardiovascular disease dataset to support ML-based risk prediction; overseeing data extraction, standardisation, and analysis of clinical records at UCH Ibadan to address data scarcity in low-resource settings.

Benefits and Significance of a Gender-Specific Multi-Specialist Diagnostic Centre in Ibadan Metropolis Nov 2024 – Present

Principal Investigator. Cross-sectional study on the role of gender-specific diagnostic centres in healthcare access and utilisation in Ibadan, assessing cultural/religious influences on patient preferences via survey and statistical analysis.

AI & HEALTH TECHNOLOGY EXPERIENCE

Lead AI Engineer Aug 2025 – Present

Empress Health AI, Silicon Valley, CA

Led design and development of an AI-driven perimenopausal healthcare platform, integrating LLM and RAG capabilities with specialist matching and personalised health-product recommendations. Built a RAG chatbot on 5,000+ clinician-vetted Q&As grounded in WHO guidelines.

Independent AI Researcher — Brain Tumour Segmentation Jan 2026 – Present

Machine Learning Collectives

Leading an independent medical imaging research project evaluating Mamba-based State Space Models against Vision Transformers and CNNs for 3D brain tumour segmentation, focused on performance, generalisability, and compute efficiency on African imaging datasets (Narval/Compute Canada HPC).

BRAINIAC — Agentic AI RAG Assistant Nov – Dec 2025

Google Kaggle Agents Intensive Capstone

Built an agentic RAG assistant powered by Gemini 2.5 Flash with Pinecone-based retrieval and automated web-search fallback; implemented multi-agent, function-calling architecture for intelligent query routing.

Team Coordinator Feb 2026 – Present

SPARK Academy

Coordinate and mentor the Oyo State cohort in AI-for-medical-imaging research, project development, and conference submissions.

CardioRisk Predictor & Teeb_2.0 AI Chatbot May – Oct 2025

SPARK Academy

Built and deployed a Streamlit cardiovascular risk predictor, improving accuracy from ~60% to ~80% via feature engineering and data augmentation (Top 9 Student Tech Innovator of the Year, 2025). Also developed a RAG pharmacology chatbot (Pinecone + LangChain + Google Generative AI).

SKILLS

LLM & Generative AI: Gemini, RAG, agentic AI, prompt engineering, LLM orchestration

Deep Learning: PyTorch, TensorFlow, CNNs, Vision Transformers, UNETR-style segmentation

HPC: Narval / Compute Canada cluster (Digital Alliance of Canada Compute Grant)

Clinical Skills: Patient clerking, clinical assessment, surgical triage

AI Infrastructure: LangChain, Pinecone, vector databases, embeddings

Programming & Tools: Python, Jupyter, Git/GitHub, Streamlit, Matplotlib

Research: Scientific writing, literature review, experimental design, data analysis

Languages: English, Yoruba, Arabic

EDUCATION

MBBS, Medicine and Surgery — 500 Level Jan 2021 – Present

College of Medicine, University of Ibadan

Clinical training at University College Hospital; passed First MB (MB1) and Second MB (MB2) professional examinations.

Artificial Intelligence and Machine Learning (Intensive, project-based) Feb – Oct 2025

SPARK Academy

Hands-on training in deep learning, computer vision, and medical imaging, applied to CNNs, feature engineering, and image segmentation for African and resource-constrained healthcare settings.

GRANTS & HONOURS

- **MICCAI Society Membership Grant** (2026) — recognition of academic and research excellence in medical image computing.
- **Digital Alliance of Canada (Narval/Compute Canada) Compute Grant** — HPC access for BraTS-Africa segmentation research.
- **Nigeria AI Collective Compute Grant** (Data Scientist Network) — HPC infrastructure for AI/medical imaging research.
- **Adaptation Lab Grant** (Apr 2026), United States.
- **Top 9 Student Tech Innovator of the Year** (UIRESDEV, 2025) — for the CardioRisk Predictor App.
- Invited Panelist, AMHS Scientific Research Conference, Uniilorin Teaching Hospital (Apr 2026) — AI in healthcare.